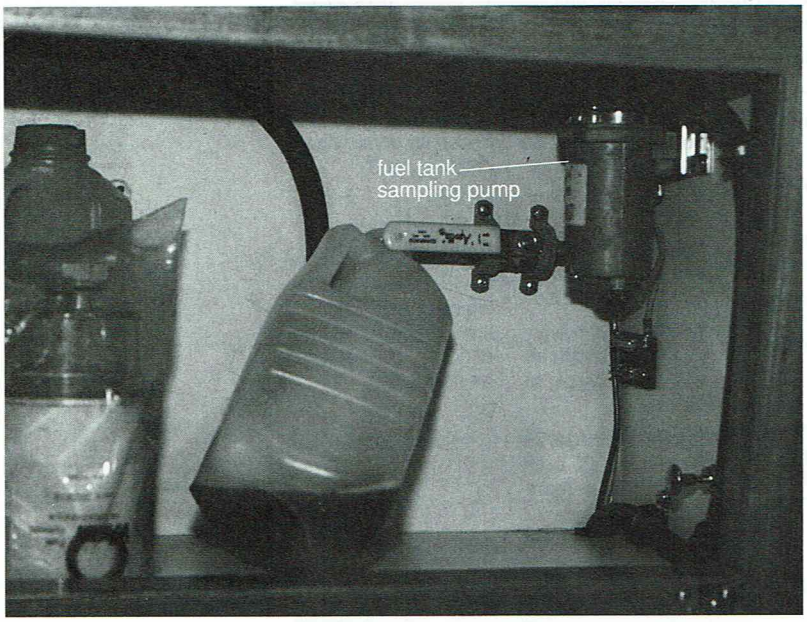


FIGURE 9-7C. Built-in fuel tank sampling pump on our old Pacific Seacraft 40.

drain the tank or pump out the fuel until no trace of contamination remains. Any especially dirty batch of fuel should be completely discarded—it's not worth risking the engine for the sake of a tankful of fuel. (On our own boats, and all others with which I am involved in the design process, I always include a fuel-sampling pump that is permanently plumbed to the lowest point in the main fuel tank—Figure 9-7C. This allows me to withdraw a fuel sample at any time and pump it into a see-through container, such as an old plastic milk jug or jam jar; all I need is 1 cup of fuel, and serious contamination is immediately apparent. *This inexpensive measure will do more to protect your fuel system than anything else you can do.* We have twice found contamination that we were able to remove before it did any damage.)

- When leaving the boat unused for long periods (e.g., when it is laid up over the winter), fill the fuel tank to the top. This eliminates the air space and cuts down on condensation in the tank. Also add a biocide. When returning the boat to service and before you crank the engine, pump a fuel sample from the base of the tank to remove any condensates.
- Periodically flush the fuel tank(s) to remove accumulated sediment.

Fuel filters. Most people regard fuel filters as the first line of defense against contaminated fuel. As should be clear by now, I regard them as the *last line of defense*, whose function is to deal with any *minor* contamination that escapes the preventive measures designed to keep the fuel tank clean.

Without exception, every marine diesel engine should have both a primary and a secondary fuel filter. All engines come from the manufacturer with an engine-mounted secondary filter located somewhere just before the fuel injection pump. *If this is the only filter, a primary filter MUST be installed* (Figures 9-8A and 9-8B). It needs to be mounted *between* the fuel tank and the lift pump, not *after* the lift pump, because any water in the fuel supply that passes through a lift pump gets broken up into small droplets that are hard to filter out.

Primary and secondary filters do not have the same function. A primary filter is the main defense against water and larger particles in the fuel supply, but it does not guard against microscopic particles of dirt and water. These are filtered out by the secondary filter.

A primary filter needs to be a *sedimenter* type that is specifically designed to separate water from fuel. Sedimenters are extremely simple, generally consisting of little more than a bowl and deflector plate. The incoming fuel hits the deflector plate, then flows around and under it to the filter outlet. Water droplets and large particles of dirt settle out. The better-quality filters then pass the fuel through a relatively coarse filter element (10 to 30 microns).

A primary filter should have a see-through bowl with a drain plug or valve so that water can be rapidly detected and removed. The ABYC

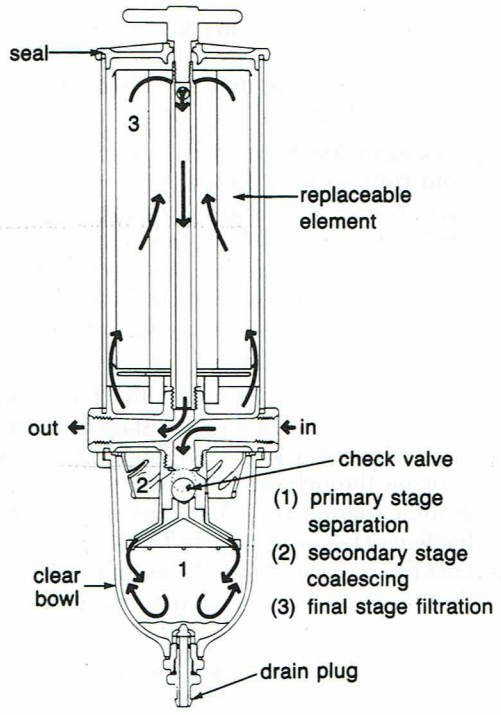


FIGURE 9-8A. A primary fuel filter with both a sedimenter function and a replaceable filter element. *(Racor)*